

Zhonghe (Catherine) Zheng

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EDUCATION

Princeton University

Sep 2025 – Present

Ph.D. Student in Mechanical and Aerospace Engineering

Advisor: Prof. Anirudha Majumdar — Research: Robot Learning, World Models

California Institute of Technology

Jun 2025

B.S. in Computer Science (Machine Learning & AI Track) with a minor in Robotics

RESEARCH EXPERIENCE

IROM Lab

Sep 2025 – Present

Advisor: Prof. Anirudha Majumdar

- Surveying action-conditioned video world models with an emphasis on controllability, long-horizon prediction, and retention of pretrained visual knowledge
- Developing a novel action-conditioned world model architecture to improve controllability while mitigating catastrophic forgetting during fine-tuning
- Fine-tuned robot policies within a world model trained on autonomously collected play trajectories and conducted real-world hardware validation experiments

Autonomous Robotics and Control Lab

Oct 2024 – Jun 2025

PI: Prof. Soon-Jo Chung Mentor: Sorina Lupu

- Worked on a custom biped platform exploring imitation learning and reinforcement learning for energy-efficient locomotion
- Trained PPO-based locomotion policies in Isaac Lab for energy-efficient bipedal walking

NASA Jet Propulsion Laboratory — Research Intern

Jun 2024 – Aug 2024

Mentor: Dr. Ashish Goel

- Validated a novel hazard avoidance algorithm Optimization Based Kinematic Settling (OBKS) for the Endurance mission
- Developed a scalable evaluation pipeline for validating hazard-avoidance algorithms across multiple simulated terrains
- Integrated OBKS into a state lattice planner and compared performance against the Nav2 planner

Autonomous Robotics and Control Lab — Research Intern

Jun 2023 – Aug 2023

PI: Prof. Soon-Jo Chung Mentors: Dr. Matt Anderson, Joshua Cho

- Led electrical integration for an autonomous drone car prototype
- Integrated emergency power systems and customized flight controller communication pipelines
- Conducted software testing and debugging while improving code modularity
- Experimented with 6-DoF controller for the platform

CHAOS — NASA Big Idea Challenge

Jan 2023 – Jun 2023

- Contributed to brainstorming, technical development, and proposal writing

PUBLICATIONS

- Yin, T., Z. Mei, **Z. Zheng**, M. Yamane, D. Wang, J. Sceats, S. M. Bateman, L. Zha, A. Badithela, O. Shorinwa, and A. Majumdar. 2026. “PlayWorld: Learning Robot World Models from Autonomous Play.” *arXiv preprint arXiv:2603.09030*.
- Mei, Z., T. Yin, O. Shorinwa, A. Badithela, **Z. Zheng**, J. Bruno, M. Bland, L. Zha, A. Hancock, J. F. Fisac, and P. Dames. 2026. “Video Generation Models in Robotics: Applications, Research Challenges, Future Directions.” *arXiv preprint arXiv:2601.07823*.
- **Zheng, C.** 2024. “Autonomous Flying Ambulance: Hardware Integration and Testing.” *Caltech Undergraduate Research Journal*.
- Singh, P. R., N. Ng, A. Patel, C. Garrido, H. Carey, S. Daghljan, R. I. Ray, K. Pulungan, S. Polidoro, **C. Zheng**, and S.-J. Chung. 2023. “SPAMMM – Solar Powered Array for Melting Materials on the Moon.” In *2023 Regional Student Conferences*. p. 71914. 2023.

TEACHING EXPERIENCE

ME/CS/EE 133A: Robotics — Teaching Assistant Fall 2024

- Held office hours and recitations covering robotics, linear algebra, and Git
- Explained concepts covering rigid body kinematics, dynamics, Jacobians, and manipulability
- Graded assignments and exams while providing detailed feedback

ME/CS/EE 129: Experimental Robotics — Teaching Assistant Spring 2024

- Mentored student teams in robotics software, ROS architectures, and hardware debugging
- Provided weekly technical guidance and code reviews

CS 21: Decidability and Tractability — Teaching Assistant Winter 2024

- Assisted students with automata theory, reductions, computability, and NP-completeness
- Held office hours and graded assignments with constructive feedback

HONORS AND AWARDS

Gordon Wu Fellowship 2025

Princeton University Graduate Fellowship

Marcella Bonsall SURF Fellowship 2023

Summer Undergraduate Research Fellowship

Edward C. Stone Voyager SURF Fellowship 2024

Summer Undergraduate Research Fellowship @ JPL

PRESENTATIONS

SURF Seminar Day Aug 2023

- Presented the Autonomous Flying Ambulance electrical integration project
- Discussed design tradeoffs, implementation challenges, and final system performance

Jet Propulsion Laboratory End-of-Project Presentation Aug 2024

- Presented Endurance project results to the JPL Endurance team
- Covered OBKS validation, hazard detection, and planner integration results

SKILLS

Programming: Python, C++ , ROS2, PyTorch, Isaac Lab, SKRL, Git

Hardware & CAD: SolidWorks, AutoDesk, 3D Printing, Soldering, Crimping, Laser Cutting

LEADERSHIP & SERVICE

Health Advocate

Sep 2024 – Jun 2025

- Completed emergency response and CPR training and earned EMR certification
- Served as a health and emergency resource within the student community

Peer Advocate

Sep 2023 – Jun 2025

- Trained in suicide prevention and peer mental health support
- Supported students experiencing emotional distress and related challenges